Chapter 5
COMPUTER INTEGRATION

NOTICE
Potential for Data Loss and/ or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- **Record data collected from procedures in this chapter into Form F4879 when directed, located in Chapter 11 of this book.**
- **Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.**

CHAPTER 5 CONTENTS

APPENDIX F CONTENTS

Section 1.0 Introduction

This chapter describes the reconfiguration, system state restore, options, and monitor adjustment procedures.

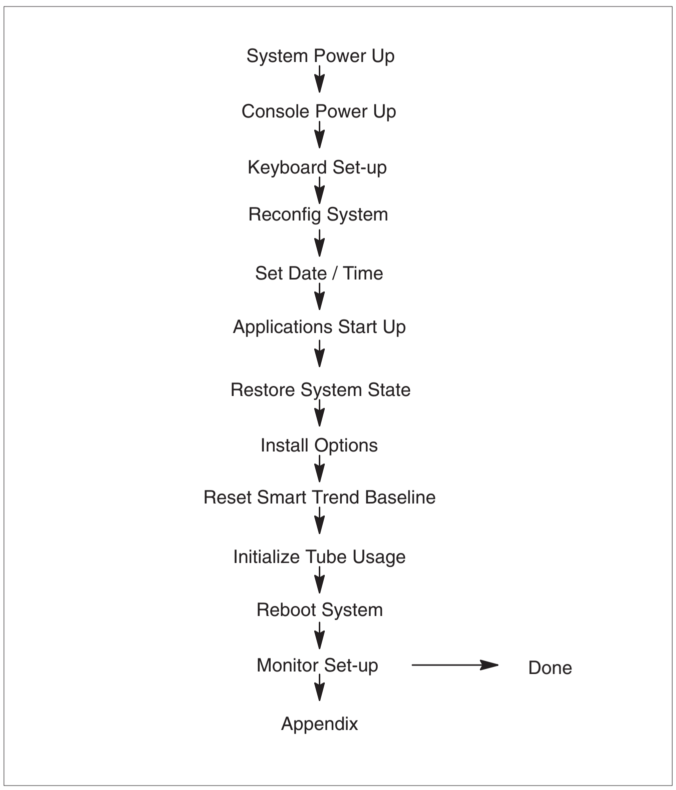

Figure 5-1 Computer Integration Process Overview

Section 2.0
System Configuration Worksheets

Complete the system configuration worksheets. Consult with your customer or network administrator to obtain the following information before you begin this section. Understanding how the customer plans to use their QX scanner, and their network and filming expectation will reduce your time to reconfigure the system.

- Manual film composer options
- IP address and net/broadcast masks
- Host ethernet address
- Networking application configuration (image transfer)
- Camera application configuration (DASM or DICOM)

2.1 Manual Film Composer Options

MANUAL FILM COMPOSER OPTIONS	
Slide Format (if available):	
Greyscale:	
Auto Printing:	
Auto Clear Page:	
Icon Labels:	
Expose Order:	
No. of Copies:	

Table 5-1 Manual Film Composer Options

2.2 System Network Configuration

SYSTEM NETWORK CONFIGURATION			
	FIELD NAME:	**SETENV NAME:**	**FIELD VALUE:**
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number *	* ask customer or checklog	
	DAS Type	DASTYPE	
	PDU Type	PDUTYPE	
Network Settings:	Gateway Host Name	GATEWAY_HOSTNAME	
	Gateway IP	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Suite Name	SUITEID	
Option	Network Printer IP Address		
Option	HIS Server IP Address		
Option	HIS Server AE Title		
Option	HIS server AE Port		
Option	CT Server AE Title		
Option	Connect Pro IP Address		

Table 5-2 System Network Configuration

2.3 Host Ethernet Address

Record your Host's Ethernet Address below:

_________:_________:_________:_________:_________:_________

2.4 Network Application (Image Transfer) Configuration

Record the network application (image transfer) configuration.

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION				
AE TITLE OR HOST NAME	NETWORK ADDRESS	NETWORK PROTOCOL	PORT NUMBER	COMMENTS

Table 5-3 Networking Application (Image transfer) Configuration

2.5 Camera Application Configuration

Record the camera application configuration for the DASM or DICOM print camera.

DASM LASER CAMERA CONFIGURATION	
Camera Type:	
DASM Type:	
Film Smooth/Sharp Setting:	
Options:	
Valid Film Formats:	
Default Film Formats:	

Table 5-4 DASM Laser Camera Configuration

DICOM PRINT CAMERA CONFIGURATION	
Camera Type:	
Host Name:	
IP Address:	
AE Title:	
TCP/IP Listen Port:	
Comments (Optional):	
Valid Film Formats:	
Default Film Formats:	
Destination:	
Orientation:	
Medium Type:	
Magnification Type:	

Table 5-5 DICOM Print Camera Configuration

DICOM PRINT CAMERA ADVANCE CONFIGURATION	
Smoothing Type:	
Configuration:	
Minimum Density:	
Maximum Density:	
Empty Density:	
Border Density:	
Association Timeout:	
Session Timeout:	
N-Set Timeout:	
N-Action Timeout:	
N-Create Timeout:	
N-Delete Timeout:	
N-Get Timeout:	

Table 5-6 DICOM Print Camera Advanced Configuration

Section 3.0
Starting the System

NOTICE Data Loss Possible **This procedure instructs you to REGEN DATABASE when you execute a reconfig. You must REGEN the DATABASE during the initial installation, or upgrade, of a system. (The initial installation or upgrade assumes the disks do not contain any calibration data.) The reconfig process, described in the following procedure, destroys any existing system calibration data.**

3.1 Power Up –> Stop for Maintenance

1.) Power up the RCU.
2.) When the following message window appears, "`Starting up the system . . .`"
3.) Select STOP FOR MAINTENANCE.

3.2 Select Keyboard Configuration

The LightSpeed System accommodates one of four keyboard configurations:

- English - US (default)
- German - DE
- Swedish - SE
- French - FR

Comment: *The computer leaves the factory configured for the English - US keyboard. If your system uses the English keyboard, continue with section 3.3.*

NON-ENGLISH KEYBOARD CONFIGURATION

A screen pops up with several icons:

- Start System
- Install System Software
- Run Diagnostics
- Recover System
- Enter Command Monitor
- Select Keyboard Layout

1.) Use the mouse to choose the Select Keyboard Layout icon.
2.) Select the appropriate keyboard type. The currently supported keyboards are:
    - English - US (default)
    - German - DE
    - Swedish - SE
    - French - FR
3.) Select APPLY to reconfigure.

3.3 Start System –> Stop the LightSpeed Auto-Start

1.) Select START SYSTEM.
2.) Select CANCEL, when the pink attention window appears, to stop Auto-Start.

Section 4.0 Reconfig

Note: Type the text shown in **boldface**, and press the **ENTER** key on the keyboard. (Type/enter `bold` = Type `bold` **ENTER**.)

4.1 Reconfig the OC

Note: When you reconfigure the OC, the software automatically reconfigures the SBC as well (if installed). The document collector box that arrived with your system contains the LightSpeed QX/i Software Load Procedure manual, which documents the reconfiguration procedure in more detail.

4.1.1 Purpose

This Reconfiguration Procedure describes how to modify the LightSpeed QX/i software configuration. This Reconfiguration allows most of the parameters entered during a LFC, to be changed without a complete LFC. The procedure consists of executing reconfig on the OC, changing the configuration on the screens, and accepting the reconfig (to allow the changes to be made).

Note: **Reconfiguration only needs to be performed on the OC (Octane Computer). In systems with an O2 Computer, the SBC (O2 Computer) reconfiguration is performed automatically in parallel with the OC reconfiguration.**

The following table indicates the parameter settings that can be changed using this Reconfiguration Procedure:

SYSTEM SETTINGS	PREFERENCES	HARDWARE SETTINGS	NETWORK SETTINGS
Hospital Name	Screen Display Language	Network Image Printer	Suite Name
Service ID	Preferred Fastcal kV		Host Name/AE Title
Time Zone	Date Display Format		Host IP Address
Next Exam #	Time Display Format		Net Mask
Scan Database Regeneration	Dose Information Display		Broadcast Address
			Default Gateway Address
			NIS (Yellow Pages)

Table 5-7 System Reconfiguration Settings

4.1.2 Procedure

On the following screens, you should make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When you are done, you can either press the ACCEPT button to start the reconfiguration process, or press the QUIT button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a shell window which closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the following file:

`/var/adm/install.log.`*`YYYMMDDWWWHHMMSS`*
(Where *`YYYYMMDDWWWHHMMSS`* is the Date/Time that the reconfiguration was started).

To view the file, type: **`more /var/adm/install.log.`*`YYYMMDDWWWHHMMSS`***

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Simply press the QUIT button at the top of the screen.
There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig entering the correct parameters.

For systems containing an O2 computer, when you reconfigure the OC, the software automatically reconfigures the SBC as well.

1.) If Applications is currently running, you must shutdown system applications to the ctuser desktop.
    a.) Click on the SERVICE DESKTOP button.
    b.) On the desktop toolbar select TOOLBOX/UTILITIES.
    c.) Select APPLICATIONS SHUTDOWN (to bring down applications only).
2.) On the OC, open a UNIX Shell window.
    a.) Type: **`su -`** **ENTER** at the prompt.
    b.) Enter the root (super user) password: **`#bigguy`**
3.) Change directory to scripts:
    a.) Type: **`cd /user/g/scripts`** **ENTER** at the prompt.
4.) Launch the LightSpeed Install utility:
    a.) Type: **`reconfig`** **ENTER** at the prompt.

    The OC displays the LightSpeed Install Utility Window as shown in Figure 5-2.

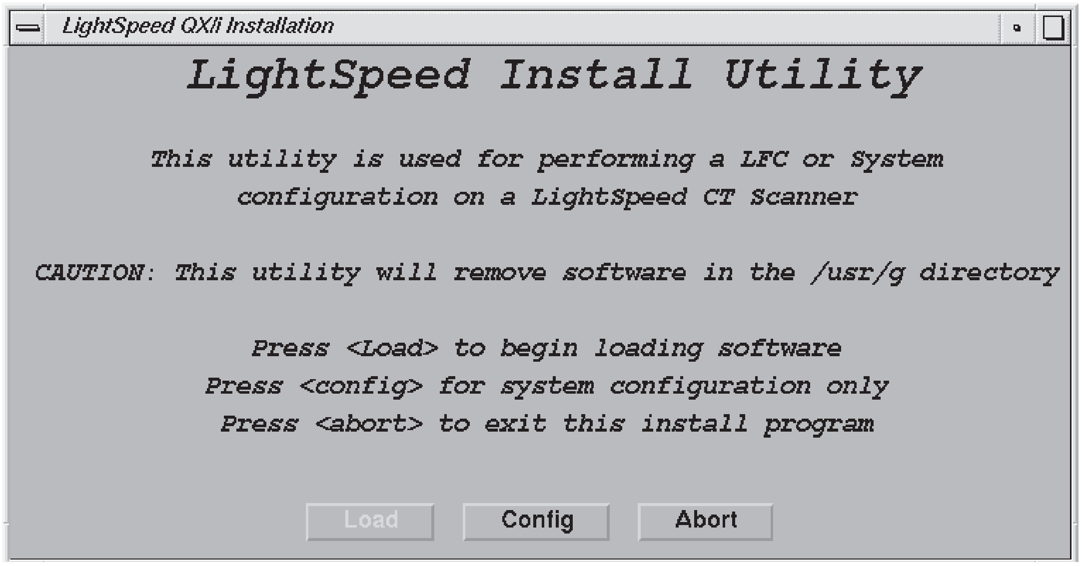

Figure 5-2 Install Utility Window

5.) Using the mouse, click on the CONFIG button.

The OC displays the LightSpeed QX/i System Configuration - System Settings Screen as shown in Figure 5-3.

Comment: *The following pages show the screens which are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens will vary depending on the current configuration of your system.*

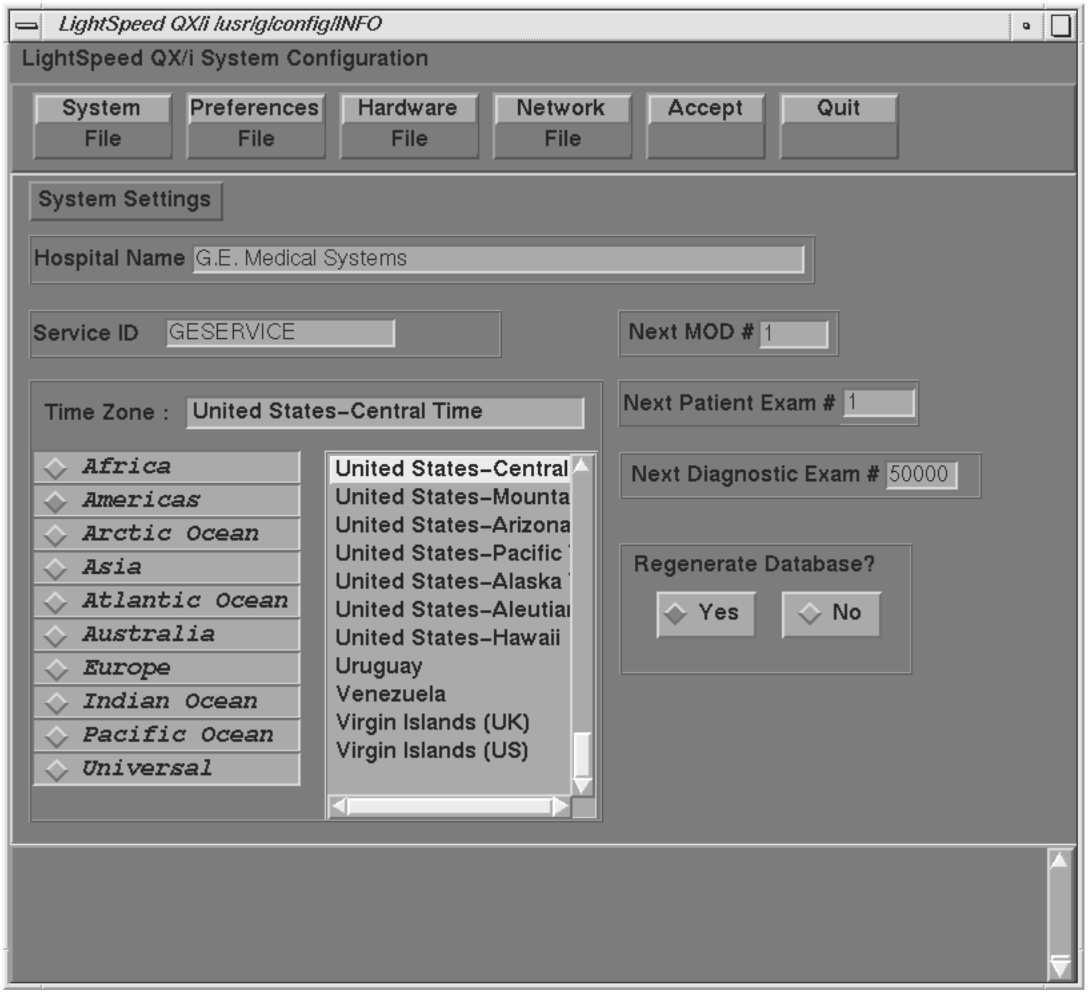

Figure 5-3 System Settings Screen

6.) `Regenerate Database` determines whether the Scan Database will be regenerated during reconfiguration.

Note: If YES is selected, then all scan data will be deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run.
Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database.

This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. See Load from Cold manual, if you require more information regarding Scan Database Regeneration.

7.) Configure System Settings

a.) `Hospital Name` configures the name that will show up on images produced by this scanner.

b.) `Service ID` is issued by the Service organization.

c.) Select the `Time Zone` for the site.

Note: Use the scrollbar at the bottom of the time-zone selection list to view the entire description of the time-zone you are about to select, to ensure that you are selecting the correct time-zone for your location.

If the time-zone of your location is not in the list above, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect. See Load from Cold manual, if you require more information regarding time-zone setting & selection.

d.) `Next MOD #` *Currently this field is not implemented.*

e.) `Next Patient Exam #` configures the next Exam number the scan user interface will use. At initial system installation, type: **1**

f.) `Next Diagnostic Exam #` *Currently this field is not implemented.*

8.) Select the PREFERENCES button to display the Preference Settings Screen as shown in Figure 5-4.

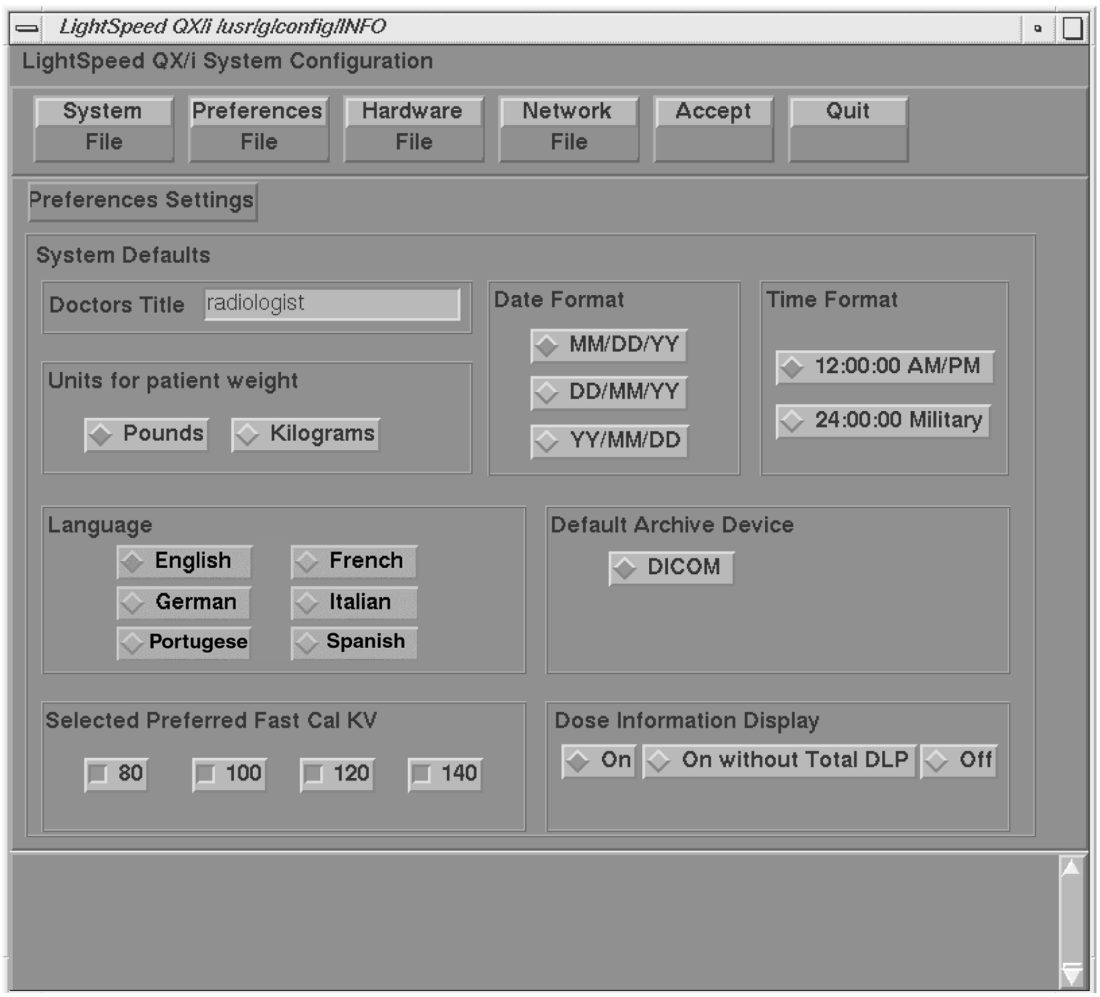

Figure 5-4 Preferences Setup Screen

Note: The `Doctor's Title` and `Units for patient weight` options are not currently implemented.

9.) Configure Preferences Settings

a.) `Language` configures the language to be displayed on the Application screens (ENGLISH, FRENCH, GERMAN, ITALIAN, PORTUGESE, SPANISH).

b.) `Preferred Fast Cal KV` configures the preferred kV that the Fast Cal Routine will calibrate (80, 100, 120, 140 in the `Selected Preferred Fast Cal KV` field). The default selections are 120 and 140.

Comment: *These kVs should include all kVs which the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs*

c.) `Date Format` and `Time Format` configures the format in which the date and time will be displayed on the images.

d.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose:

* Select ON (full CTDiw Display)
* Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
* Select OFF (no CTDIw Display)

10.) Select the HARDWARE button to display the Hardware Settings Screen as shown in Figure 5-5.

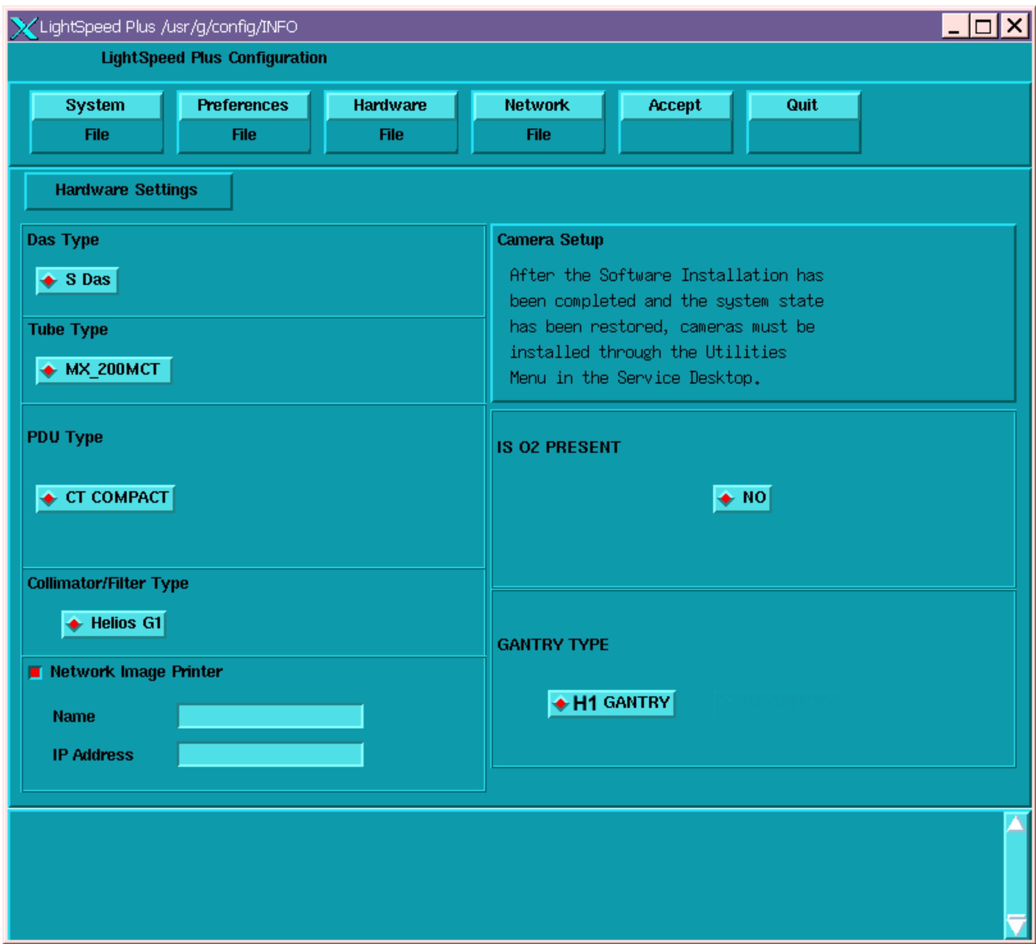

Figure 5-5 Hardware Settings Screen

11.) Configure Hardware Settings

a.) `DAS Type` configures which DAS the software should expect. Presently, LightSpeed systems only support S DAS.

b.) `Tube Type` configures which X-Ray Tube the software should expect. Presently, LightSpeed systems only support MX_200MCT.

c.) `PDU Type` configures which PDU the software should expect. Presently, LightSpeed systems only support CT COMPACT.

d.) `Collimator/Filter Type` configures which Collimator/Filter the software should expect. Presently, QX/i systems only support HELIOS G1.

e.) Verify the presence or absence of an O2 computer in your console, and select YES or NO as appropriate. The default setting for this selection comes from a value maintained in the system's INFO file.

Note: The O2 configuration selection can only be made during software load (not reconfig). If you have incorrectly made the wrong selection during LFC, you must restart LFC from the beginning to correct this.

f.) `Gantry Type` selects the type of Gantry connected to this console. Currently, QX/i systems only support H1 GANTRY for Gantry Type.

g.) Select Network Image Printer only if the site plans to use a network connected postscript printer (such as Codonics printers) with this system. When selected:

* Enter the printer Hostname in the `Name` field.
* Enter the printer IP Address in the `IP Address` field.

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e.: a computer with a printer attached); a Network Image Printer (which is supported) is a printer directly attached to the network.

h.) Camera Setup:

Comment: *Camera Setup is not performed using Reconfig, but rather is performed from the Service Desktop once Applications have been started.* *See "Declaring DICOM Filming Devices on the LightSpeed System" on page 288.*

12.) Select the NETWORK button to display the Network Settings Screen as shown in Figure 5-6.

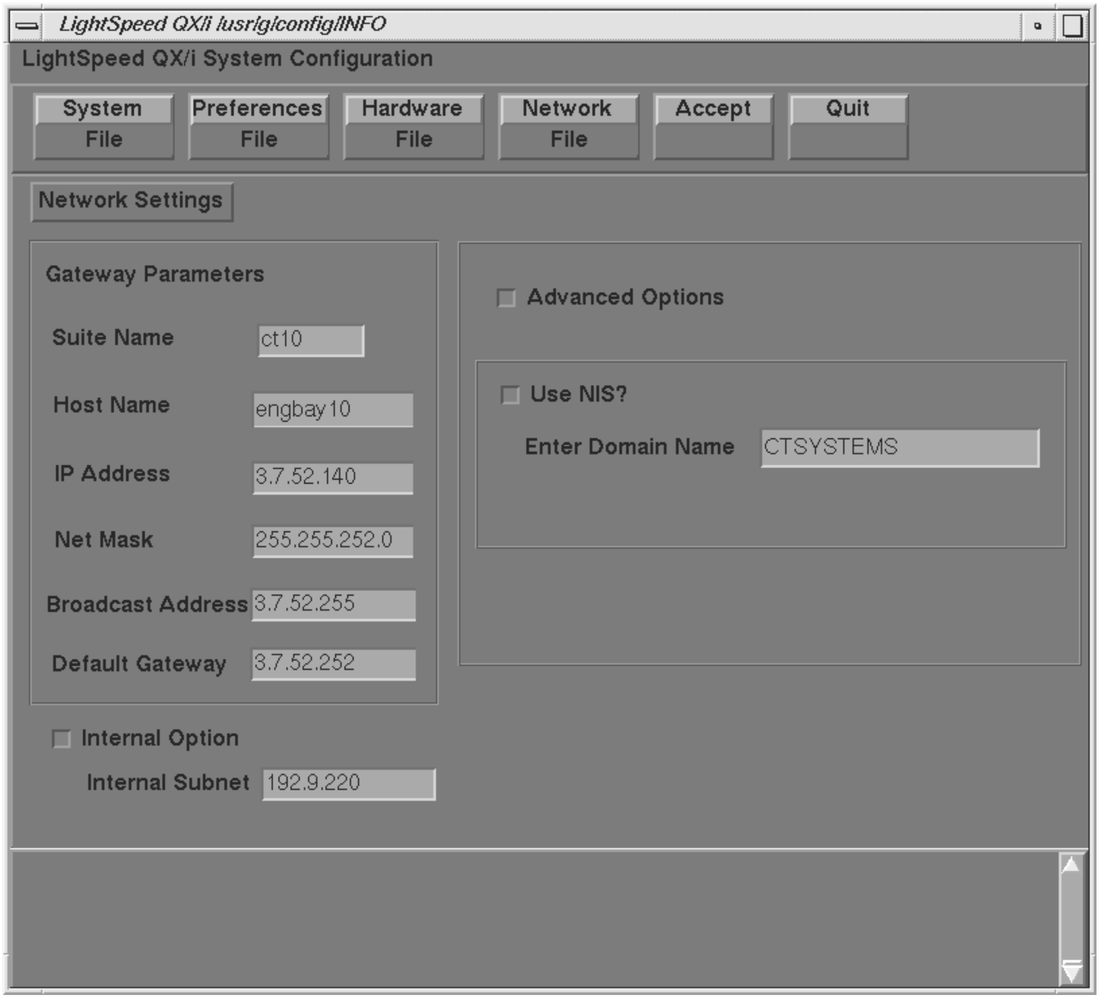

Figure 5-6 Network Settings Screen

13.) Configure Network Settings

Comment: *This screen provides the ability to declare the LightSpeed system on a hospital network. Key information such as Host Name, IP Address, Net Mask, and Default Gateway must be obtained from the hospital's network administrator.*

a.) Enter the `Suite Name`.

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters. Total must be four characters long. The name of the OC interface will be *`<Suite Name>`*`_oc` within the scanner's subnet, and the SBC interface will be *`<Suite Name>`*`_sbc` (for systems that contain an O2 Computer) within the scanner's subnet. Typically you should use CT01, unless the customer prefers a different Suite Name.

b.) Enter the hospital provided `Host Name`.

Note: The Host Name identifies the network hostname and AE Title of the LightSpeed system to the hospital's network.

The Host Name**:**

* **MUST NOT** be *<Suite Name>*_oc or *<SUITE NAME>_OC*.
* **MUST NOT** exceed 16 Characters.
* **MUST** only contain the following characters: **A through Z, a through z, 0 through 9, - and _**

c.) Enter the hospital provided `IP Address` for the system.

d.) Enter the hospital provided `Net Mask` (if the LightSpeed system is on a subnet).

e.) Enter the `Broadcast Address`.

Comment: The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1's or 0's depending on the configuration of the network. The standard default is 1's, but older Sun OS machines used O's.

For example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.

If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

---

f.) Enter the hospital provided `Default Gateway` IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.

g.) Select NIS (a.k.a. Yellow Pages database) `Advanced Option` only if requested by the hospital network administrator as follows:

* Select ADVANCED OPTIONS button on the Network Settings Screen.
* Select USE NIS button.
* Enter the hospital provided `Domain Name`.

14.) Select ACCEPT on the LightSpeed QX/I System Configuration Screen.

Comment: The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC (and for systems containing an O2 computer, the SBC).

The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window which closes when the loading process is complete. All the window output is logged to a file named:

`/var/adm/install.log.YYYYMMDDWWWHHMMSS`

`(Where` *YYYYMMDDWWWHHMMSS* is the Date/Time that the loading process was started).

15.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See Figure 5-7.)

Figure 5-7 Reboot Screen

16.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.

APPLICATIONS START-UP

17.) In the Console shell, type: `st` ENTER

4.2 Check/Set Date and Time

Check the date and time.

1.) Open a UNIX Shell on the OC.
2.) Type/enter `su - root` **ENTER** to change to root.
3.) Type/enter `setdate` **ENTER**.
4.) Follow the instructions to set the correct time and date.
5.) Verify the dates and times on the OC and the SBC are the same. If the dates/times do not match, refer to the Software Load Procedure manual for more information regarding time zone selection and time setting.
6.) Type `exit` **ENTER**.

4.3 Applications Start-Up

Open the Console shell window, and type: `startup` **ENTER**. The applications desktop appears on the monitor.

4.4 QX/i Service Desktop Main Menu

The service desktop (Figure 5-8) is the entry point for all service tools and diagnostics. The desktop is designed with nine major functional menu areas each with its own purpose. These areas are:

- Error Logs - Select and review system logs.
- Diagnostics - Select and execute all diagnostic applications.
- Image Quality Tools - Image quality tools not requiring communications via firmware with the system (such as image analysis and scan analysis).
- Calibration Applications - Tools for mechanical, electrical, and imaging calibrations of the

system.

- Configuration Applications - Save/restore system state and configuration information.
- Replacement Parts/Repair Procedures - Links to tools required when replacing major field replaceable units (FRUs).
- Toolbox/Utilities - Useful to the field engineer while installing or servicing a system.
- Planned/Preventive/Proactive Maintenance - Information to execute a PM visit.
- Service Desktop Home Page - Icon descriptions and eventually system health status information.

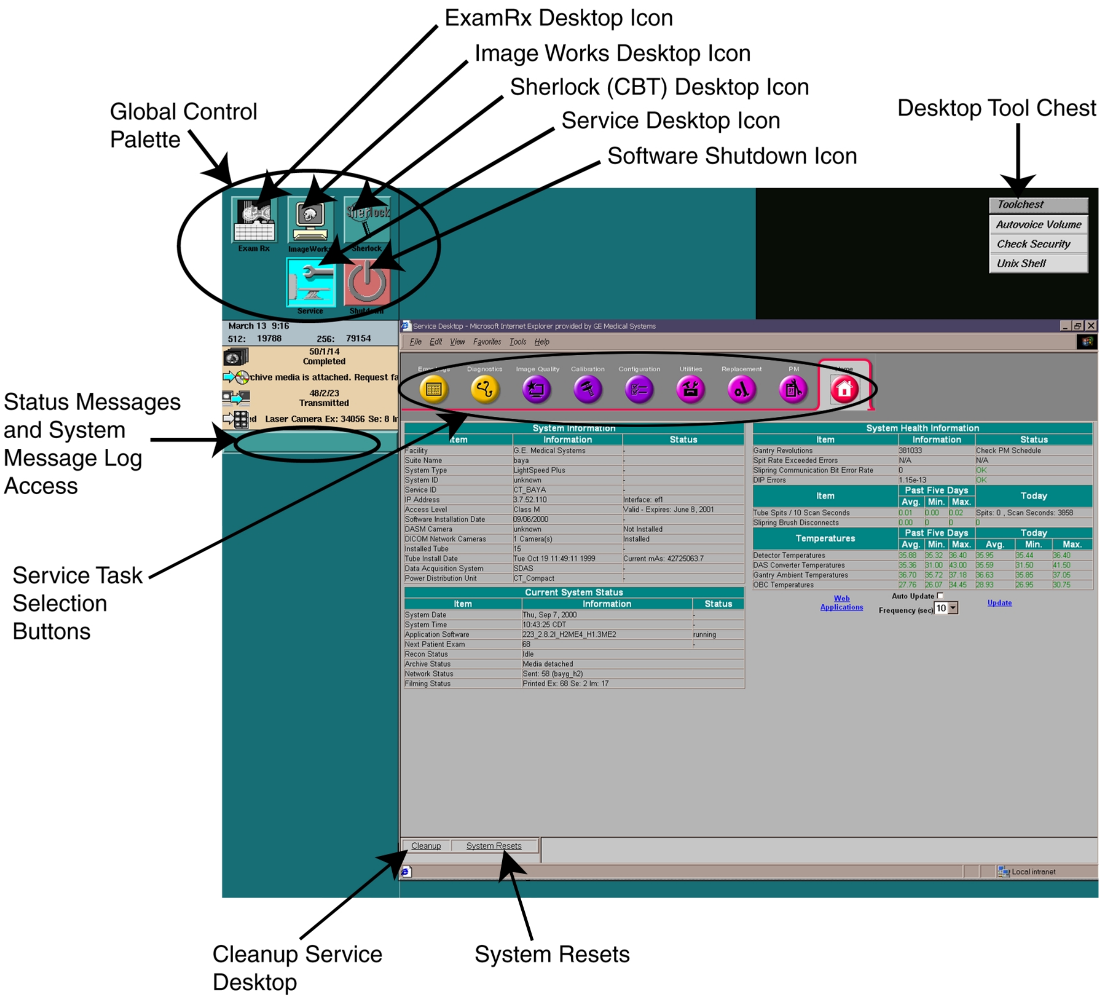

Figure 5-8 Service Desktop, Display Screen Overview

4.5 Toolbox/Utility Install Sub-Menu (Overview)

The icon sequence used to access the toolbox/utility install sub-menu is as follows:

Figure 5-9 Service Desktop Icon

Figure 5-10 Toolbox/Utilities Icon

The toolbox/utility install sub-menu (Figure 5-11) appears. This menu provides a single access point on the service desktop to work from when integrating and testing a newly installed system prior to turn over to the user.

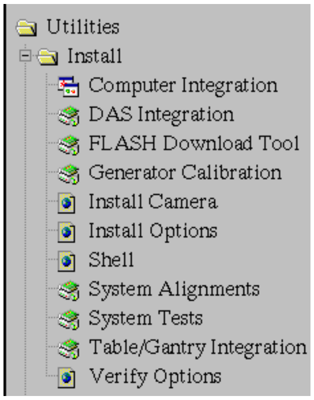

Figure 5-11 Toolbox/Utility Install Submenu

The menu has three types of icons:

1.) The first icon  represents tools that require the download of diagnostic firmware to the scan control sub-system.

2.) The second icon  represents tools that require the download of application firmware to the scan control sub-system.

3.) The third icon  represents tools that do not require the download of any firmware to the scan control sub-system.

A particular tool is executed by clicking on its icon or on the text next to the icon.

Note: **Brackets surrounding a name on the menu indicates it is a planned feature, one not yet implemented.**

4.6 Configure Camera

See Chapter 12 for more information and complete details of setting up a DICOM Print Camera.

1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
2.) Click on the TOOLBOX/UTILITIES icon.
3.) Click on INSTALL.
4.) Click on INSTALL CAMERA.
5.) Refer to the Software Load Procedure manual, for details on setting up the Camera Configuration. There is troubleshooting information in Chapter 3 of the System Service Manual.

Section 5.0
Restore System State

Your system should have a system state MOD, located in the software collector box.
The system state MOD contains:

- Collimator Characterization

- Phantom Calibrations

The installation process uses all the system state files. At this time, use the system state MOD to restore the phantom calibrations and system characterization files.

If you cannot locate an existing system state MOD, follow instructions in Appendix G on page 187 to manually characterize the system and perform phantom calibrations.

1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
2.) Click on the PM icon. 
3.) Select SYSTEM STATE.
4.) Insert the MOD in the MOD drive.
5.) Select CHARACT.
6.) Select CALS.
7.) Select RESTORE to restore the system characterization and phantom calibration files to the system. If any error should occur during the restore process, see the Software Load Procedure manual for information regarding possible error messages and their recovery.

Note: Restore State can take as long as ten minutes, although typical times average about three minutes. When Restore State completes, dismiss the tool, and proceed to the next section.

8.) Click NO for `Reset Scan Hardware` popup message.
9.) Select DISMISS.

Section 6.0 Install Customer Options

Note: **Your system may have a MOD that contains customer purchased options. If your system has an options MOD, install it at this time.**

Ensure that the options MOD is NOT write protected at this time. The initial install requires that the MOD be write enabled; subsequent installs can be done with the MOD write protected.

1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
2.) Click on the TOOLBOX/UTILITIES icon.
3.) Select INSTALL.
4.) Select COMPUTER INTEGRATION.
5.) Select INSTALL OPTIONS.

An Options Window appears (Figure 5-12):

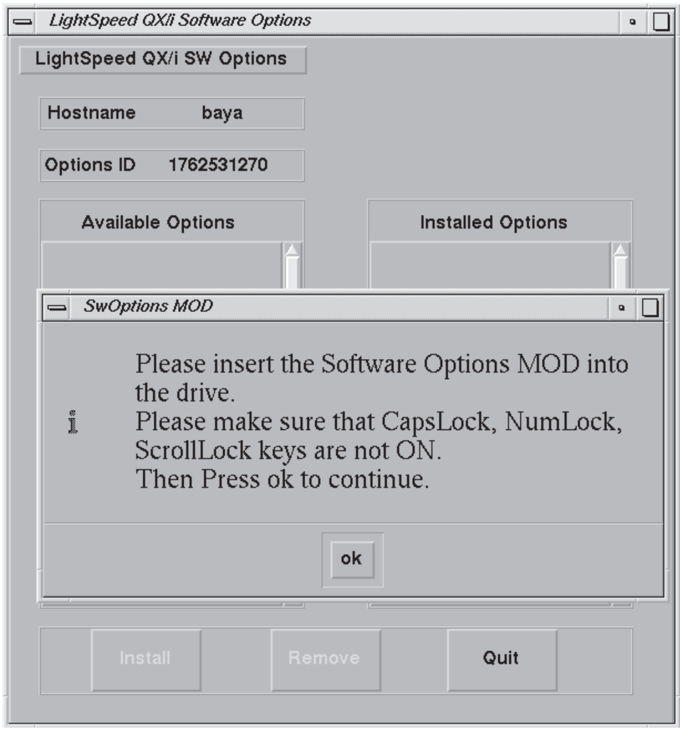

Figure 5-12 Options Window when First Selected

6.) Insert the options MOD into the MOD drive and click on OK. (If you do not have an options MOD, click on OK anyway, wait for the abort pop up, then abort the process.)

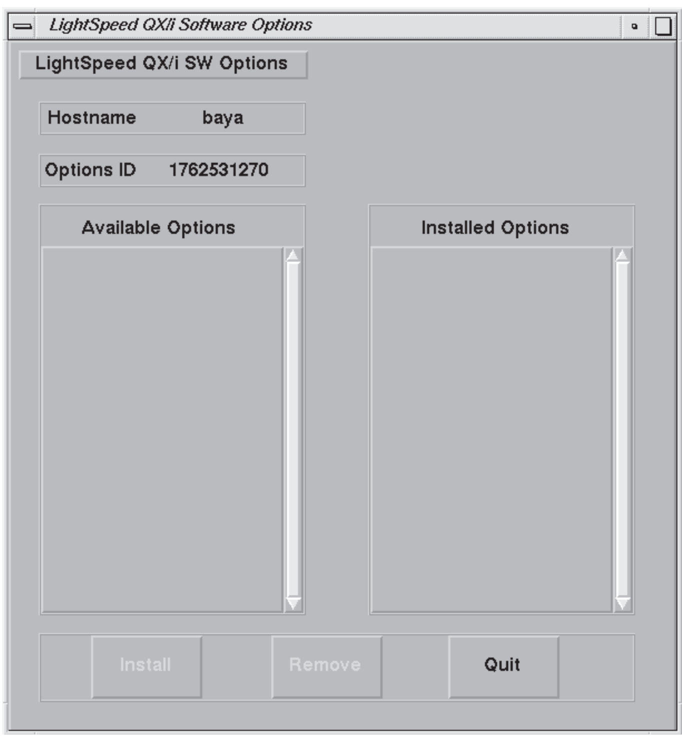

Figure 5-13 Example: Options Window

7.) Select all of the options in the left-hand column to install the corresponding software.

8.) Select INSTALL**.** A box may appear while the options are loading. When an option is displayed in the `Installed Options` list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half minute (due to the fact that they are installing software).
9.) After the options are installed, select QUIT.
10.) Select OK.
11.) Remove the MOD and write protect the side with options.
12.) When the system prompts to Reboot, click YES, and reboot the system to complete the installation.

Section 7.0 Initialize Tube Usage

Performing this procedure is important to ensure that the customer's tube usage for the scanner starts with the system installation (and that scans that may have been taken during the manufacturing process do not get included in their usage record).

1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
2.) Click on the TOOLBOX/UTILITIES icon.
3.) Select TOOLS.
4.) Select TUBE USAGE to open the New Tube Config tool.
5.) Answer the questions. The system will ask for Tube Insert Serial Number and Tube Housing Serial Number.
6.) Open the Gantry and look for the rating plate located on the tube housing.
7.) Enter the Serial Numbers from this rating plate into the system.

Note: Shutdown and start-up the system to update the tube usage count to the correct value.

Section 8.0 Entering Detector ID

1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
2.) Click on the TOOLBOX/UTILITIES icon.
3.) Select INSTALL.
4.) Select COMPUTER INTEGRATION.
5.) Select ENTER DETECTOR ID.
6.) Open the gantry and look for rating plate/bar code located on the end of the detector.
7.) Enter the serial number of the detector.

Section 9.0 Monitor Set Up

This section describes the steps required to set up the two display monitors. The flowchart in Figure 5-14 provides an overview of the monitor setup procedure. Use the detailed procedures in the following sections to set up the monitors.

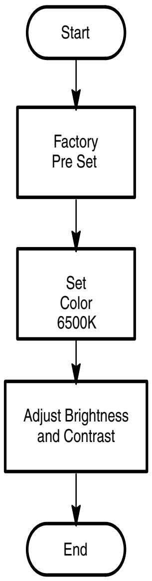

Figure 5-14 Monitor Setup Flowchart

9.1 Factory Preset

Refer to Section 3.0 on page 169 of Appendix F for the monitor's basic setup procedure.

9.2 Set the Color Temperature

Refer to Section 4.0 on page 171 of Appendix F for the procedure to set the color temperature to 6500K.

9.3 Monitor Brightness and Contrast Setup

Color monitors have a lower light output than black and white monitors. Take care when you adjust the monitor brightness & contrast on LightSpeed systems.

If the operators tell you they think the image on the monitor looks **softer** than the image on the film, *and* they like the film, adjust the monitor brightness and contrast to compensate for its lower light output. Use the following procedure to adjust the monitor brightness and contrast, until the operator sees the anatomical structure (window width) at the expected window levels.

1.) Adjust the ambient room light to the level used by the operator during image review.

2.) Use one of the following methods:

a.) Click the IMAGEWORKS icon 

and proceed to step 3.

OR

b.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.

c.) Click on the IMAGE QUALITY icon. 

d.) Select INSTALL SMPTE IMAGE and proceed to step 3.

3.) Display the SMPTE pattern. Use the browser to select Exam 1000, which contains the SMPTE pattern, and enlarge the image to full screen display.

4.) Increase the Contrast to maximum.

5.) Increase the Brightness to maximum.

6.) Decrease the Brightness, until the raster just fades into, and matches, the monitor screen background. At this point, the 5% and 95% patches should be just visible.

If additional tweaking is required to attempt to match the monitor image to the filmed image, use only the brightness control.

If the CRT image exhibits any tearing or smearing of the alphanumeric characters, then reduce the contrast setting slightly until the tearing/smearing is just eliminated. The optimum setting for contrast is the highest setting that does not cause tearing/smearing of the alphanumeric characters.

You should always finish up by displaying and filming images of anatomy (typical heads and bodies), and asking the technologist to compare the CRT image to the film image.

9.4 Customizing the Monitor

Refer to Section 4.0 on page 171 of Appendix F for procedures to adjust the size, centering, and convergence of the Sony monitor.